Shuping Li

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EDUCATION

- **Ph.D. in Global Hydrology** — Tokyo, Japan
  Department of Civil Engineering, The University of Tokyo — 2020.09 – 2024.09
  Advisor: Prof. Dai Yamazaki
- **M.E. in Hydraulic Engineering** — Guangzhou, China
  School of Geography and Planning, Sun Yat-sen University — 2017.09 – 2019.07
  Advisor: Prof. Guoping Tang
- **B.E. in Hydrology and Water Resources Engineering** — Guangzhou, China
  School of Geography and Planning, Sun Yat-sen University — 2013.09 – 2017.07
  Advisor: Prof. Guoping Tang

WORK EXPERIENCES

- **Project Researcher (Postdoctoral level)** — Tokyo, Japan
  Institute of Industrial Science, The University of Tokyo — 2024.10 – Present
- **Research Assistant** — Tokyo, Japan
  Department of Civil Engineering, The University of Tokyo — 2023.10 – 2024.09
- **Research Intern** — Guangzhou, China
  South China Institute of Environmental Sciences — 2019.07 – 2020.08
- **Visiting Scholar** — New Jersey, US
  Civil and Environmental Engineering, Princeton University — 2026.02
  Department of Earth and Planetary Sciences, Rutgers University-New Brunswick — 2023.03

RESEARCH INTERESTS

- Hydrological modeling at hillslope scale in land surface model
- Impact of hillslope water dynamics on land cover heterogeneity
- Interaction between soil moisture and vegetation dynamics
- Land-atmosphere interaction

PUBLICATIONS

M. First author and corresponding (*) author publications:

[M6] **Resolving Climate Heterogeneity along Hillslope in Land Surface Model**
<u>Li, S.*</u> et al.
In preparation

[M5] **Kilometer-scale simulation systematically improves land–atmosphere coupling estimates**
<u>Li, S</u>.*, Tokuda, D., Hsu, H., Shih, C., Hsu, J., Lo, M., Takasuka, D., Tanoue, M., Yoshimura, K.
Under review

[M4] **Resolving Land Cover Heterogeneity along Hillslope Improves Simulation of Terrestrial Water and Energy Budgets**
<u>Li, S.*</u>, Yamazaki, D., Tozawa, T., Adachi, K., Nitta, T., Zhao, G., Zhou, X., Yoshimura, K.
Water Resources Research, 61(9), 2025

[M3] **Where in the World Are Vegetation Patterns Controlled by Hillslope Water Dynamics?**
<u>Li, S.*</u>, Yamazaki, D., Zhou, X., Zhao, G.
Water Resources Research, 60(4), 2024

[M2] **Soil moisture-vegetation interaction from near-global in-situ soil moisture measurements**
<u>Li, S.*</u> and Sawada, Y.
Environmental Research Letters, 17(11), 2022

[M1] **Concentration estimation of dissolved oxygen in Pearl River Basin using input variable selection and machine learning techniques**
Li, W., Fang, H., Qin, G., Tan, X., Huang, Z., Zeng, F., Du, H.* and **<u>Li, S.*</u>**
Science of the Total Environment, 731, 139099, 2020

C. Other collaborative publications:

[C2] **Developing a Levee Module for Global Flood Modelling with a Reach-Level Parameterization Approach**
Zhao, G.*, Yamazaki, D., Tanaka, Y., Zhou, X., **<u>Li, S.</u>**, Hu, Y., Hirabayashi, Y., Neal, J.C., Bates, P.D.
Water Resources Research, 61(8), 2025

[C1] **Streamflow response to snow regime shift associated with climate variability in four mountain watersheds in the US Great Basin**
Tang, G.*, **<u>Li, S.</u>**, Yang, M., Xu, Z., Liu, Y., Gu, H.
Journal of Hydrology, 573(March), 255–266, 2019

BOOKS

[B1] **Description of MATSIRO6**
MATSIRO6 document writing team, Guo, Q., Kino, K., **<u>Li, S.</u>**, Nitta, T., Takeshima, A., Suzuki, K. T., et al.
Runoff (Chapter 9) and Tile scheme (Chapter 13), 2021

CONFERENCE PRESENTATIONS

O. Oral presentations:

[O10] **AOGS Annual meeting 2026** — Fukuoka, Japan, 2026.08
Revisiting Land-Atmosphere Coupling Across Spatial Scales: From Coarse to Kilometer-Scale Simulations

[O9] **EGU General Assembly 2026** — Vienna, Austria, 2026.05
Revisiting Land-Atmosphere Coupling Across Spatial Scales: From Coarse to Kilometer-Scale Simulations

[O8] **EALSA Workshop** — Jeju, South Korea, 2026.01
Revisiting Land-Atmosphere Coupling Across Spatial Scales: From Coarse to Kilometer-Scale Simulations

[O7] **CAS-TWAS-WMO forum** — Beijing, China, 2025.10
Resolving Land Cover Heterogeneity Along Hillslope Improves Simulation of Terrestrial Water and Energy Budgets

[O6] **AOGS Annual meeting 2025** — Singapore, 2025.07
Resolving Land Cover Heterogeneity Along Hillslope Improves Simulation of Terrestrial Water and Energy Budgets

[O5] **JSHWR-JAHS conference 2024** — Tokyo, Japan, 2024.09
Resolving land heterogeneity in land surface model modulates terrestrial water and energy budgets

[O4] **9th GEWEX Conference** — Sapporo, Japan, 2024.07
Representing hillslope-scale land surface heterogeneity in land surface model substantially modulates water and energy budget

[O3] **AGU Fall meeting 2022** — Chicago, US, 2022.12
Where can we find unique landscapes regulated by hillslope moisture dynamics?

[O2] **JSHWR-JAHS conference 2022** Kyoto, Japan
An efficient method to succinctly represent the explicit land cover heterogeneity formed by hillslope water dynamics 2022.09

[O1] **JpGU Annual meeting 2022** Chiba, Japan
An efficient method to succinctly represent the explicit land cover heterogeneity formed by hillslope water dynamics 2022.05

P. Poster presentation:

[P5] **Land Surface Model Summit 2** Rome, Italy
Resolving Climate Heterogeneity along Hillslope in Land Surface Model 2026.09

[P4] **Hydro-JULES event** London, UK
Resolving Land Cover Heterogeneity Along Hillslope Improves Simulation of Terrestrial Water and Energy Budgets 2025.09

[P3] **JpGU Annual meeting 2025** Chiba, Japan
Resolving Land Cover Heterogeneity Along Hillslope Improves Simulation of Terrestrial Water and Energy Budgets 2025.05

[P2] **AGU Fall meeting 2024** Washington, D.C., US
Resolving Hillslope Heterogeneities in Land Surface Model 2024.12

[P1] **AGU Fall meeting 2021** New Orleans, US
Soil moisture-vegetation correlation from 733 in-situ soil moisture observations 2021.12

INVITED TALKS

[T4] **National Institute of Natural Hazards** Beijing, China
Resolving Land Cover Heterogeneity along Hillslope Improves Simulation of Terrestrial Water and Energy Budgets 2025.10

[T3] **IGSNRR, Chinese Academy of Science, Forum for Water Problem** Beijing, China
Resolving Land Cover Heterogeneity along Hillslope Improves Simulation of Terrestrial Water and Energy Budgets 2025.10

[T2] **UKCEH-UTokyo, Global Hydrology seminar series** Online
Resolving Land Cover Heterogeneity along Hillslope Improves Simulation of Terrestrial Water and Energy Budgets 2025.03

[T1] **Dept. of Hydraulic Engineering, Tsinghua University** Online
Soil moisture-vegetation correlation from 3239 in-situ soil moisture observations 2022.01

GRANTS AND AWARDS

- **Kurata Grant** 2025.03 – 2026.03
  The Hitachi Global Foundation (39 recipients out of 342 applicants)
- **Conference Travel support** 2024.07
  9th GEWEX Open Science Conference
- **Musha Shugyo Travel Grant** 2023.03
  School of Engineering, The University of Tokyo
- **Grant for Dispatch to International Research Meetings** 2022.12
  Foundation for the Promotion of Industrial Science (FPIS)
- **Best Presentation in HSP2021 of Hydrosphere Environmental Group** 2021.07
  Department of Civil Engineering, The University of Tokyo
- **MEXT scholarship** 2020.09 – 2023.09
  Ministry of Education, Culture, Sports, Science and Technology of Japan

MEDIA REPORTS

[R2] **Avoiding static land surface models: improvements in simulating water–energy–vegetation dynamics** 2025.09
Institute of Industrial Science, The University of Tokyo

[R1] **Finding Where the Grass is Greener** 2024.05
Institute of Industrial Science, The University of Tokyo

TEACHING EXPERIENCE

- **Global hydrodynamics and its impact on ecosystem (guest lecture)** 2026.10
  Tokyo City University

STUDENTS MENTORED

- **Siyuan Huang** UTokyo Master of Civil Engineering 2026.04 – 2026.08
- **Jules Barthes** University of Lyon, Visiting Master at UTokyo 2026.04 – 2026.09
- **Yuze Wang** UTokyo Master of GSFS → CUHK Ph.D. 2025.04 – Present
- **Shun Wakamatsu** UTokyo Master of Civil Engineering 2024.04 – 2026.03

ACADEMIC SERVICES

Reviewer for journal

- *Agroforestry Systems, CATENA, Environmental Research Letters, Earth System Dynamics, Frontiers in Plant Science, Journal of Advances in Modeling Earth Systems, Journal of Hydrology, Measurement Science and Technology, Scientific Reports,* etc.

Convener for conference

- **EGU General Assembly 2026** Vienna, Austria
  HS10.5/BG3**:** Ecohydrological responses to droughts in a changing environment: Mechanism, trends and impacts. 2026.05
- **6th Hydro90 meeting** Online
  Mechanism and evolution of ecohydrological process 2026.05

Organizer for meeting and workshop

- **ILS model developer meeting** Chiba, Japan
  2026.03
- **ILS model Couplethon workshop** Chiba, Japan
  2026.03